

FashionSales

October 26, 2025

0.1 E Commerce Fashion sales Data cleaning, Analysis & Automation

About the Dataset — Women's Fashion Products

This dataset contains 30,000 women's fashion products, providing a diverse range of apparel and accessories.

Categories Covered

Western Wear Indian Wear Perfumes & Fragrances Watches Nightwear

Column Descriptions

Column Name	Description
BrandName	The brand or manufacturer of the product.
Details	Descriptive details about the product (style, material, design, etc.).
Size	Available sizes for the product (e.g., S, M, L, XL).
MRP	Maximum Retail Price of the product (before discount).
SellPrice	The actual selling price after applying discounts.
Category	The main product category (e.g., Western Wear, Watches).

In this project we are going to Perform

Data cleaning Data Analysis Data Visualisation Automation

Libraries to be used

NumPy Pandas Matplotlib Seaborn

```
[74]: ## Importing important libraries  
import numpy as np  
import pandas as pd  
import matplotlib.pyplot as plt  
import seaborn as sns
```

```
[75]: ## Loading the dataset  
df = pd.read_csv("FashionDataset.csv")
```

```
[76]: df.head()
```

```
[76]: Unnamed: 0      BrandName      Deatils \
0      0      life  solid cotton blend collar neck womens a-line d...
1      1      only  polyester peter pan collar womens blouson dres...
2      2      fratini  solid polyester blend wide neck womens regular...
3      3      zink london  stripes polyester sweetheart neck womens dress...
4      4      life  regular fit regular length denim womens jeans ...

      Sizes      MRP SellPrice Discount \
0 Size:Large,Medium,Small,X-Large,X-Small Rs\n1699      849 50% off
1      Size:34,36,38,40 Rs\n3499      2449 30% off
2      Size:Large,X-Large,XX-Large Rs\n1199      599 50% off
3      Size:Large,Medium,Small,X-Large Rs\n2299      1379 40% off
4      Size:26,28,30,32,34,36 Rs\n1699      849 50% off

      Category
0 Westernwear-Women
1 Westernwear-Women
2 Westernwear-Women
3 Westernwear-Women
4 Westernwear-Women
```

```
[77]: df.columns
```

```
[77]: Index(['Unnamed: 0', 'BrandName', 'Deatils', 'Sizes', 'MRP', 'SellPrice',
          'Discount', 'Category'],
          dtype='object')
```

```
[78]: ## Renaming each column properly
df = df.rename(columns={'Unnamed: 0':"Sales_id", 'BrandName':"Brand", 'Deatils':
↪ "details",
          'Sizes':"Sizes", 'MRP':"MRP", 'SellPrice':"Selling price",
          'Discount':"Discount in %", 'Category':"Category"})
```

```
[79]: df.columns
```

```
[79]: Index(['Sales_id', 'Brand', 'details', 'Sizes', 'MRP', 'Selling price',
          'Discount in %', 'Category'],
          dtype='object')
```

```
[80]: ## Ordering columns in the right order
right_Order = ["Sales_id","details","Category","Sizes","Brand","MRP","Selling_
↪ price","Discount in %"]
df = df[right_Order]
```

```
[81]: df
```

```
[81]: Sales_id                                details \
0      0 solid cotton blend collar neck womens a-line d...
1      1 polyester peter pan collar womens blouson dres...
2      2 solid polyester blend wide neck womens regular...
3      3 stripes polyester sweetheart neck womens dress...
4      4 regular fit regular length denim womens jeans ...
...
30753  21 crystal stylish womens rodhium earrings
30754  22 Nan
30755  23 ethnic gold plated jhumki earrings
30756  24 womens gold plated double line fancy white and...
30757  25 womens gold plated bridge designer mangalsutra...

Category                                     Sizes \
0      Westernwear-Women Size:Large,Medium,Small,X-Large,X-Small
1      Westernwear-Women Size:34,36,38,40
2      Westernwear-Women Size:Large,X-Large,XX-Large
3      Westernwear-Women Size:Large,Medium,Small,X-Large
4      Westernwear-Women Size:26,28,30,32,34,36
...
30753  Jewellery-Women Nan
30754  Jewellery-Women Nan
30755  Jewellery-Women Nan
30756  Jewellery-Women Nan
30757  Jewellery-Women Nan

Brand      MRP      Selling price      Discount in %
0      life      Rs\n1699      849      50% off
1      only      Rs\n3499      2449      30% off
2      fratini      Rs\n1199      599      50% off
3      zink london      Rs\n2299      1379      40% off
4      life      Rs\n1699      849      50% off
...
30753  swarovski      Nan      8950      Nan
30754      Nan      Nan      Nan      Nan
30755  jewelz      Rs\n1839      643      65% off
30756  estelle      Nan      2799      Nan
30757  estelle      Nan      1899      Nan
```

[30758 rows x 8 columns]

```
[82]: ## extracting MRP from MRP column
df["MRP"] = df["MRP"].str.replace("Rs\n", "")
```

```
[83]: ## extracting discount from Discount in % column
df["Discount in %"] = df["Discount in %"].str.replace("% off", "")
```

```
[84]: df["Category"].value_counts()
```

```
[84]: Category
Westernwear-Women      10374
Indianwear-Women       10374
Lingerie&Nightwear-Women  3354
Footwear-Women         2574
Watches-Women          1794
Jewellery-Women        1794
Fragrance-Women         494
Name: count, dtype: int64
```

```
[85]: ## Removing -Women String from category column
df["Category"] = df["Category"].str.replace("-Women", "")
```

```
[86]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30758 entries, 0 to 30757
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Sales_id              30758 non-null  int64
1   details               30758 non-null  object
2   Category              30758 non-null  object
3   Sizes                 30758 non-null  object
4   Brand                 30758 non-null  object
5   MRP                   30758 non-null  object
6   Selling price         30758 non-null  object
7   Discount in %         30758 non-null  object
dtypes: int64(1), object(7)
memory usage: 1.9+ MB
```

Here, It is showing there are no null values because the null values are present as string values "Nan". we need to replace them with Numpy NaN

```
[87]: df = df.replace("Nan", np.nan)
```

```
[88]: df
```

```
[88]:      Sales_id      details \
0         0  solid cotton blend collar neck womens a-line d...
1         1  polyester peter pan collar womens blouson dres...
2         2  solid polyester blend wide neck womens regular...
3         3  stripes polyester sweetheart neck womens dress...
4         4  regular fit regular length denim womens jeans ...
...      ...      ...
30753    21  crystal stylish womens rodhium earrings
```

30754	22		NaN
30755	23	ethnic gold plated jhumki earrings	
30756	24	womens gold plated double line fancy white and...	
30757	25	womens gold plated bridge designer mangalsutra...	

	Category	Sizes	Brand \
0	Westernwear	Size:Large,Medium,Small,X-Large,X-Small	life
1	Westernwear	Size:34,36,38,40	only
2	Westernwear	Size:Large,X-Large,XX-Large	fratini
3	Westernwear	Size:Large,Medium,Small,X-Large	zink london
4	Westernwear	Size:26,28,30,32,34,36	life
...	...	...	...
30753	Jewellery	NaN	swarovski
30754	Jewellery	NaN	NaN
30755	Jewellery	NaN	jewelz
30756	Jewellery	NaN	estelle
30757	Jewellery	NaN	estelle

	MRP	Selling price	Discount in %
0	1699	849	50
1	3499	2449	30
2	1199	599	50
3	2299	1379	40
4	1699	849	50
...	...	...	...
30753	NaN	8950	NaN
30754	NaN	NaN	NaN
30755	1839	643	65
30756	NaN	2799	NaN
30757	NaN	1899	NaN

[30758 rows x 8 columns]

```
[89]: ## converting MRP, Selling price & discount columns as Float

df[["MRP","Selling price","Discount in %"]] = df[["MRP","Selling_
price","Discount in %"]].\
astype("float")
```

```
[90]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30758 entries, 0 to 30757
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Sales_id        30758 non-null  int64
1   details         29575 non-null  object
```

```

2   Category      30758 non-null object
3   Sizes         23257 non-null object
4   Brand         29575 non-null object
5   MRP           22550 non-null float64
6   Selling price 29575 non-null float64
7   Discount in % 22550 non-null float64
dtypes: float64(3), int64(1), object(4)
memory usage: 1.9+ MB

```

```
[91]: ## Checking for duplicate records
df.duplicated().sum() ## There are 1331 duplicate records
```

```
[91]: np.int64(1331)
```

```
[92]: ## Dropping duplicate records
df.drop_duplicates(inplace=True)
```

```
[93]: ## Checking for null values

df.isna().sum()
```

```
[93]: Sales_id      0
      details      7
      Category     0
      Sizes       6201
      Brand        7
      MRP         7014
      Selling price 7
      Discount in % 7014
      dtype: int64
```

```
[94]: df["details"].value_counts()
```

```
[94]: details
synthetic womens casual sandals - black
35
alloy metal gold color western womens earrings
27
solid zari silk blend festive wear womens regular saree - black
24
structured zari silk blend festive wear womens regular saree - green
24
structured zari silk blend festive wear womens regular saree - pink
23
..
the wing woman clover charm necklace
1
sterling silver oval turquoise ear studs
```

```

1
womens pearl golden drops
1
nine pieces combo fashion non-precious metal base earrings
1
womens metallic gold statement multilayered chain with bar and ring lock detail
western necklace - multi      1
Name: count, Length: 23876, dtype: int64

```

```
[95]: df["Sizes"].value_counts()
```

```

[95]: Sizes
Size:Large,Medium,Small,X-Large,X-Small      1970
Size:Large,Medium,Small,X-Large              1615
Size:Large,Medium,Small,X-Large,X-Small,XX-Large  1505
Size:Large,Medium,Small,X-Large,XX-Large      1441
Size:Large,Medium,Small,X-Large,XX-Large,XXX-Large  738
...
Size:35,36,37,38,40                          1
Size:2,3,4,6                                1
Size:5,6,11                                  1
Size:3,4,6                                  1
Size:7.5,8.5                                1
Name: count, Length: 1171, dtype: int64

```

```
[96]: df["Brand"].value_counts()
```

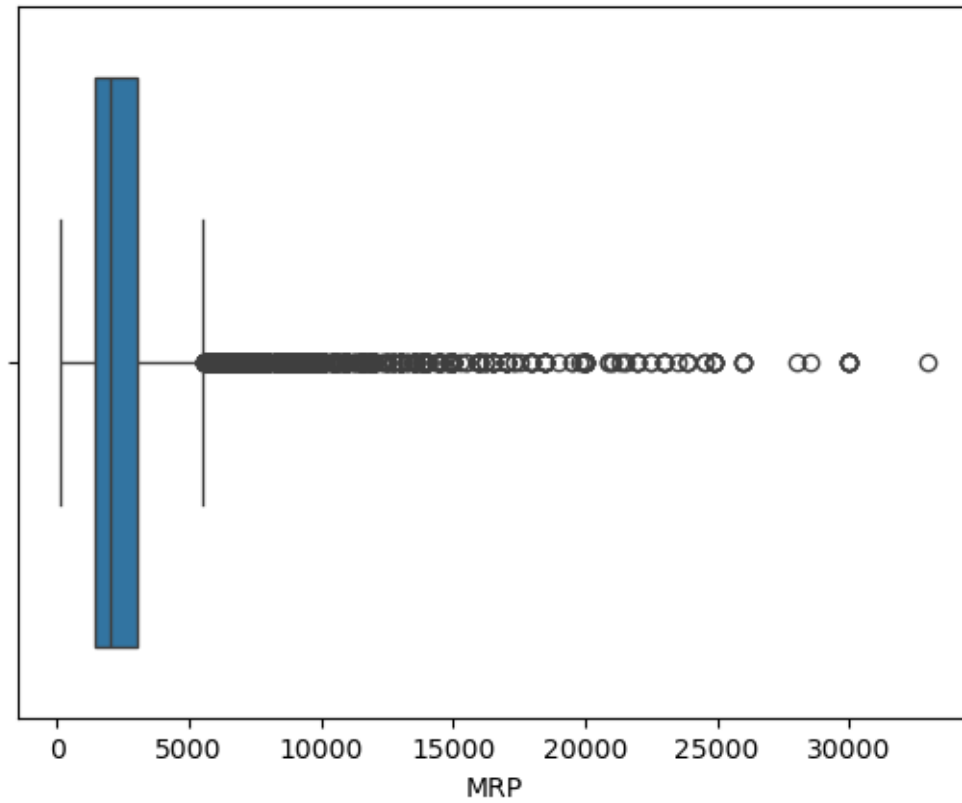
```

[96]: Brand
vastranand      1467
zink london     917
global desi     889
stop            879
enamor          789
...
dkny            1
ungaro          1
18.21 man made  1
the real effect london  1
accessorize     1
Name: count, Length: 274, dtype: int64

```

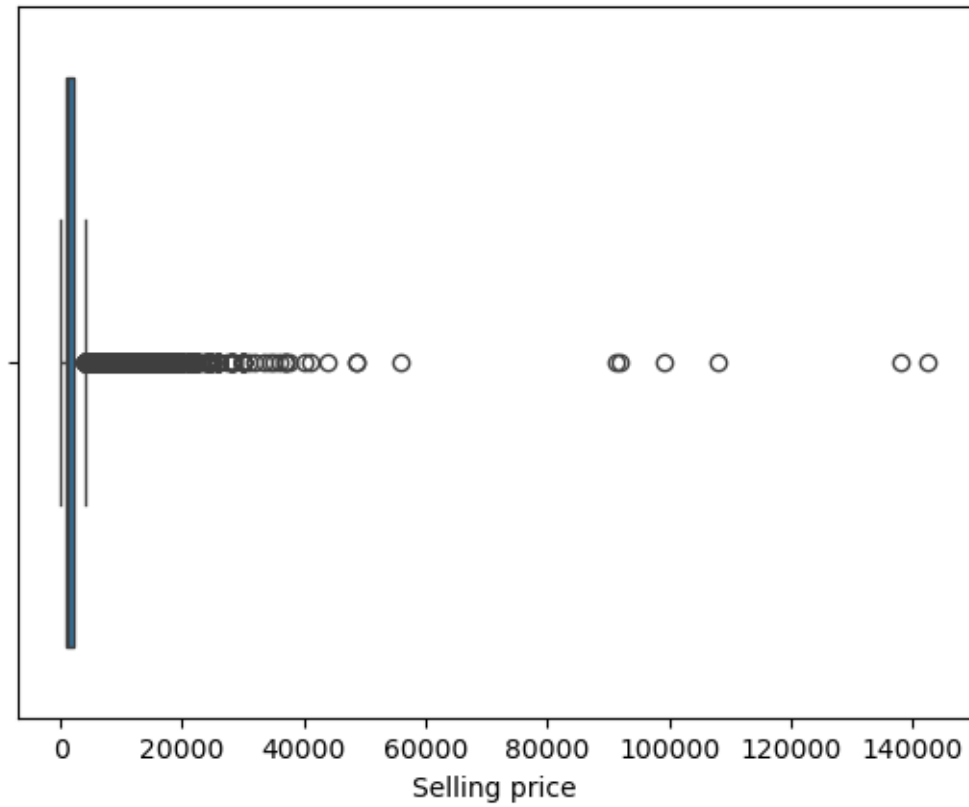
```
[97]: sns.boxplot(data=df,x="MRP")
```

```
[97]: <Axes: xlabel='MRP'>
```



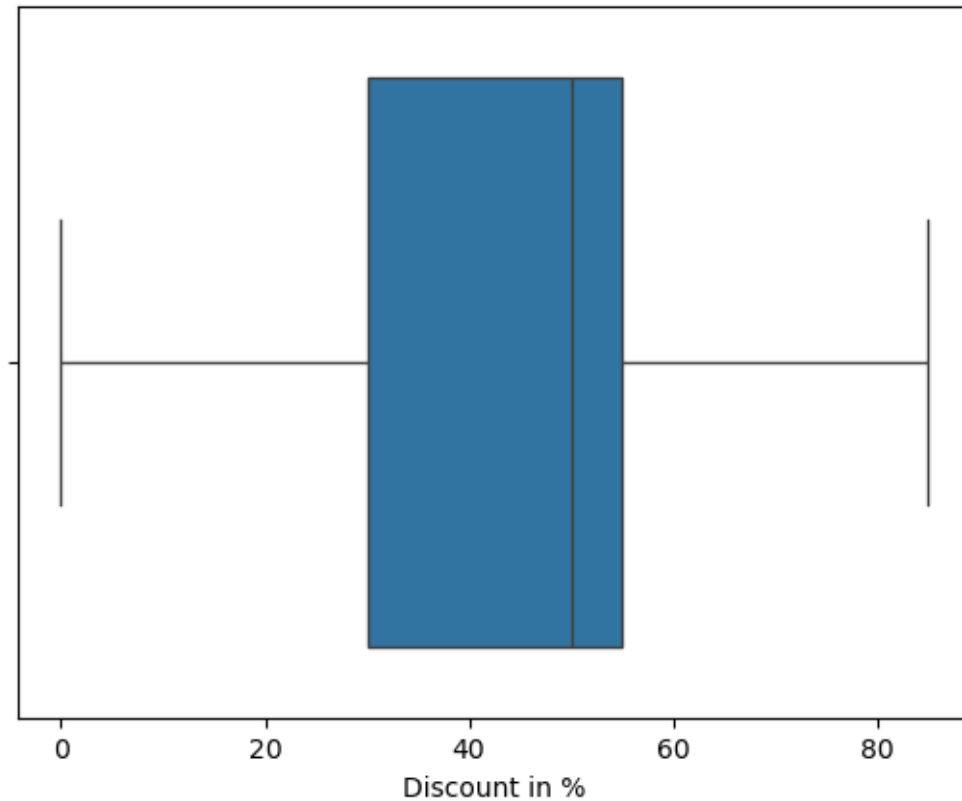
```
[98]: sns.boxplot(data=df,x="Selling price")
```

```
[98]: <Axes: xlabel='Selling price'>
```



```
[99]: sns.boxplot(data=df,x="Discount in %")
```

```
[99]: <Axes: xlabel='Discount in %'>
```



Here, details column can be dropped because it has no much significance. Sizes column can be dropped because there is no consistency in Size categories. In Brand column null values can be replaced by Mode. In Discount column null values can be replaced by Mean as it has no outliers the MRP & Selling price null values can be replaced by median since they have high number of outliers

```
[100]: ## Dropping details & sizes column
```

```
df = df.drop(columns=["details", "Sizes"])
```

```
[101]: ## replacing Brand column null values with mode
```

```
df["Brand"].mode()
```

```
[101]: 0    vastranand  
Name: Brand, dtype: object
```

```
[102]: df["Brand"] = df["Brand"].fillna("vastranand")
```

```
[103]: ## replacing Discount column null values with mean
```

```
df["Discount in %"] = df["Discount in %"].fillna(df["Discount in %"].mean())
```

```
[104]: ## replacing MRP column null values with Median
df["MRP"] = df["MRP"].fillna(df["MRP"].mean())

[105]: ## replacing Selling price column null values with Median
df["Selling price"] = df["Selling price"].fillna(df["Selling price"].mean())

[106]: df.isna().sum() ## Now, there are no more null values
```

```
[106]: Sales_id      0
       Category     0
       Brand        0
       MRP          0
       Selling price 0
       Discount in % 0
       dtype: int64
```

0.2 Category Analysis

```
[107]: ## Top categories by number of products sold
df["Category"].value_counts()
```

```
[107]: Category
       Westernwear      9962
       Indianwear      9858
       Lingerie&Nightwear 3222
       Footwear        2466
       Watches         1725
       Jewellery        1718
       Fragrance        476
       Name: count, dtype: int64
```

Westernwear, Indianwear are the categories with highest number of products sold, 9962 & 9858 respectively covering 2/3 rd of total products sold

```
[108]: ## Top categories with highest total sales
Category_sales = df.groupby("Category")["Selling price"].sum()
Category_sales = pd.DataFrame(Category_sales)
Category_sales["Selling price"] = round(Category_sales["Selling price"], 2)
Category_sales.sort_values(by="Selling price", ascending=False)
```

```
[108]:
```

Category	Selling price
Watches	14928748.34
Indianwear	14592062.34
Westernwear	13135903.34
Footwear	5420364.34
Jewellery	3731077.34
Lingerie&Nightwear	2960242.34

Fragrance 1888657.34

Watches has the sales of 1.5 Crore with just 1725 products

```
[109]: ## Categories with highest & Lowest Avg discount
Category_discount = df.pivot_table(index="Category", values="Discount in %",
                                     aggfunc="mean")
Category_discount.sort_values(by="Discount in %", ascending=False)
```

```
[109]: Discount in %
Category
Indianwear      49.765004
Westernwear     43.444641
Lingerie&Nightwear 42.926231
Jewellery       41.697450
Footwear        41.255835
Watches         39.257212
Fragrance       35.611246
```

Indianwear has the highest Avg discount of 49.6% Fragrance has the highest Avg discount of 36%

```
[110]: ## Category with costliest product
Category_max_price = df.pivot_table(index="Category", values="Selling price",
                                     aggfunc="max")
Category_max_price.sort_values(by="Selling price", ascending=False)
```

```
[110]: Selling price
Category
Watches      142500.0
Jewellery    31000.0
Fragrance    19800.0
Indianwear   13599.0
Footwear     10999.0
Westernwear   8400.0
Lingerie&Nightwear 3799.0
```

Watches has the costliest product with the price of 1.4 Lakh

0.3 Brand Analysis

```
[111]: ## Brands with highest sales
Top_selling_brand = df.pivot_table(index="Brand", values="Selling price",
                                     aggfunc="sum")
Top_selling_brand["Selling price"] = round(Top_selling_brand["Selling price"], 2)
Top_selling_brand.sort_values(by="Selling price", ascending=False)
```

```
[111]: Selling price
Brand
vastranand      3214190.4
```

michael kors	2119319.0
fossil	1550869.0
global desi	1447961.0
titan	1436843.0
...	...
gas	624.0
black panther	598.0
beverly hills polo club	500.0
ariana	400.0
unknown by ayesha	359.0

[274 rows x 1 columns]

vastranand - Indian wear brand has highest sales - 32 Lakhs

```
[112]: ## Brands with costliest products
Costly_brand = df.pivot_table(index="Brand", values="Selling_
    price", aggfunc="max")
Costly_brand.sort_values(by="Selling price", ascending=False).head(3)
```

```
[112]:          Selling price
Brand
versace      142500.0
tissot        48500.0
michael kors  40995.0
```

versace, tissot, michael kors are the Watch brand with costliest items

```
[113]: ## Brands with highest Avg discount
Brand_discount = df.pivot_table(index="Brand", values="Discount in_
    %", aggfunc="mean")
Brand_discount.sort_values(by="Discount in %", ascending=False).head()
```

```
[113]:          Discount in %
Brand
gas      75.000000
myshka   72.060519
varkala silk sarees  70.923077
unknown by ayesha   70.000000
jewelz    65.000000
```

gas - an indian wear brand has highest Avg Discount

```
[114]: df["Category"].unique()
```

```
[114]: array(['Westernwear', 'Indianwear', 'Lingerie&Nightwear', 'Footwear',
    'Watches', 'Fragrance', 'Jewellery'], dtype=object)
```

We need to export the data of each Category into separate files

```
[ ]: ## Creating dataframes for each category

Westernwear_data = df[df["Category"]=="Westernwear"]
Indianwear_data = df[df["Category"]=="Indianwear"]
Lingerie_Nightwear_data = df[df["Category"]=="Lingerie&Nightwear"]
Footwear_data = df[df["Category"]=="Footwear"]
Watches_data = df[df["Category"]=="Watches"]
Fragrance_data = df[df["Category"]=="Fragrance"]
Jewellery_data = df[df["Category"]=="Jewellery"]
```

```
[ ]: ## Exporting Data of each category

Westernwear_data.to_csv("Westernwear_data.csv")
Indianwear_data.to_csv("Indianwear_data.csv")
Lingerie_Nightwear_data.to_csv("Lingerie_Nightwear_data.csv")
Footwear_data.to_csv("Footwear_data.csv")
Watches_data.to_csv("Watches_data.csv")
Fragrance_data.to_csv("Fragrance_data.csv")
Jewellery_data.to_csv("Jewellery_data.csv")
```

0.4 Thank you